EPIDEMIOLOGICAL SKILLS

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LEARNING OBJECTIVES

- To understand clinical data in the context of broader epidemiology
- To understand 'exposure' and what it may mean for different groups
- To understand prevalence and incidence, and derive these for a clinical group

FORMAT

PART I

- What is epidemiology?
- What does it involve?
- How does sports (injury) epidemiology fit?
- Challenges and opportunities

PART II

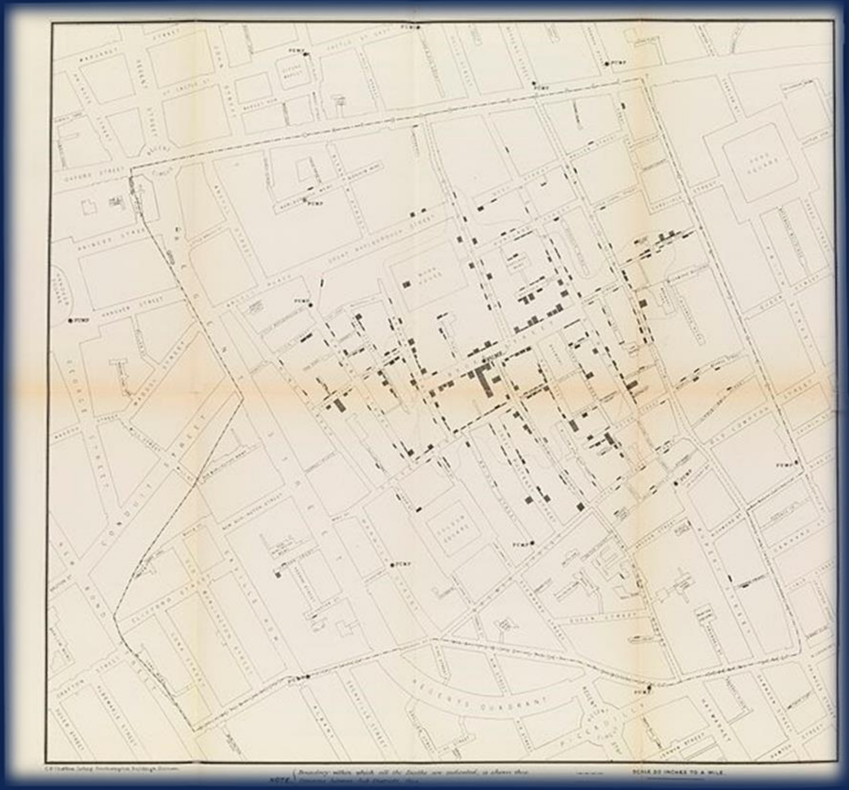
- Key concepts
- Metrics
- Cautions
- Hands-on data

EPIDEMIOLOGY

- Distribution & determinants
- Who has it? Who gets it? Why?
- What do they have in common? What sets them apart
- Sequalae: so what? How does this affect broader society? How does it affect the economy?

EPIDEMIOLOGY- MODERN ORIGINS

- Westminster, London
- Multiple cholera outbreaks (1832, 1848-1849, 1853, 1854)
- There was little understanding of hygiene, and germs were thought to be spread by miasma ('bad air')
- Dr John Snow investigated the cases and deaths, representing the location of every death by a bar, and identifying a clustering around one water pump



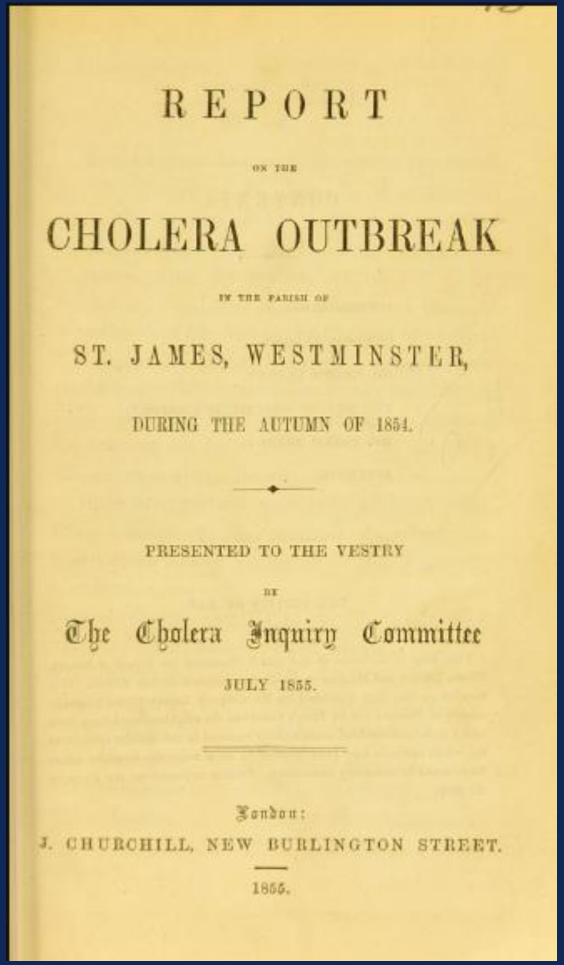
EPIDEMIOLOGY- MODERN ORIGINS

- This breakdown of who had the condition, what they had in common, and how they differed was presented to the Cholera Inquiry Committee, July 1855

> *I was informed by this lady's son that she had not been in the neighbourhood of Broad Street for many months. A cart went from broad Street to West End every day and it was the custom to take out a large bottle of the water from the pump in Broad Street, as she preferred it. The water was taken on Thursday 31st August., and she drank of it in the evening, and also on Friday. She was seized with cholera on the evening of the latter day, and died on Saturday*

- The Committee was able to discuss what similarities and differences there were – and most interestingly, those who were atypical cases – what did they share with those that were typical

- Having identified the pump as a potential cause, we can remove or disable this – and evaluate the success of our intervention


REPORT

ON THE

CHOLERA OUTBREAK

IN THE PARISH OF

ST. JAMES, WESTMINSTER,

DURING THE AUTUMN OF 1854.

PRESENTED TO THE VESTRY

BY

The Cholera Inquiry Committee

JULY 1855.

London:

J. CHURCHILL, NEW BURLINGTON STREET.

1855.

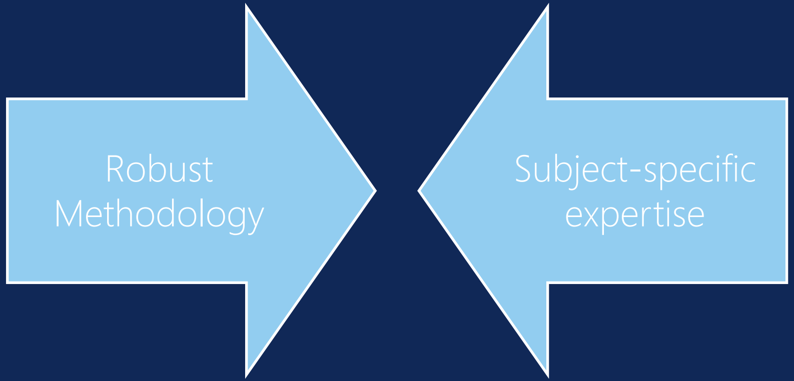
PRESENT DAY: EPIDEMIOLOGY

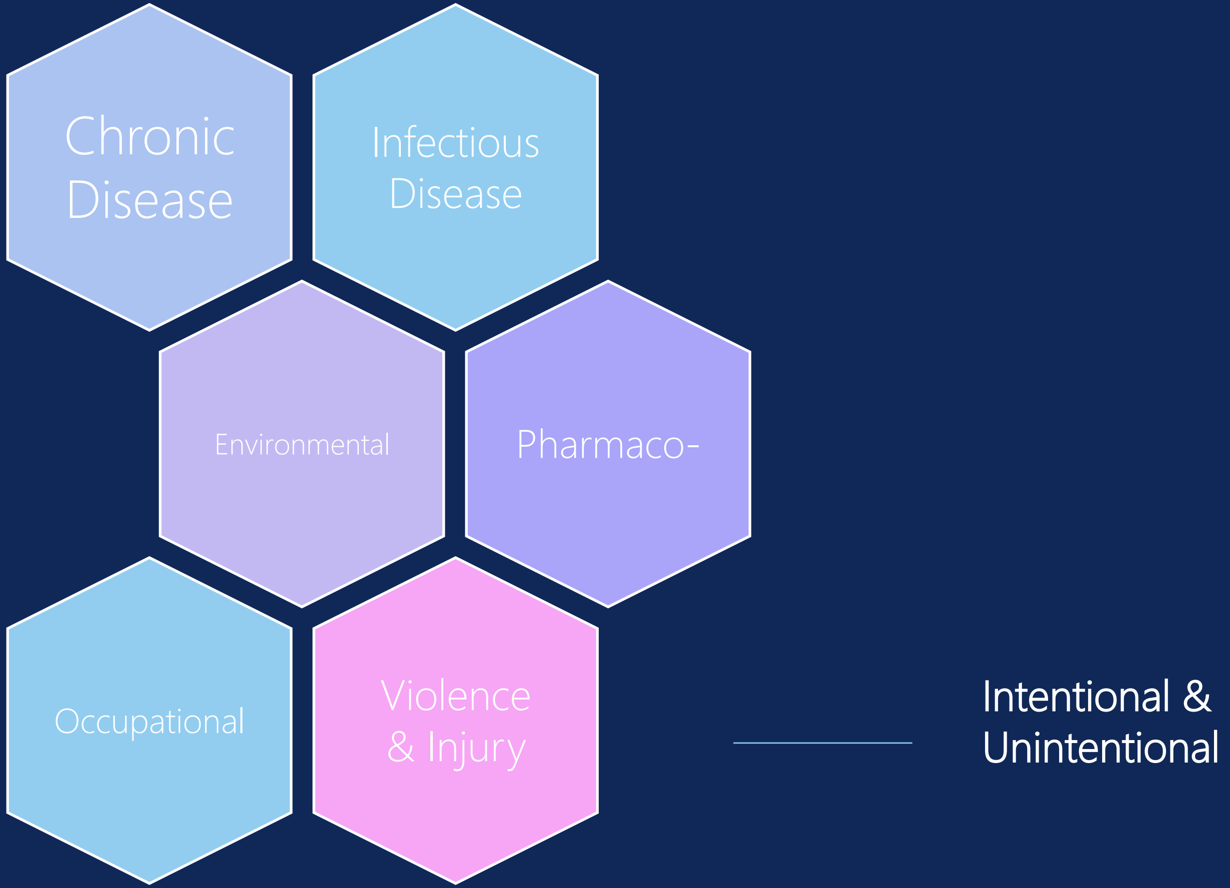
Quantitative methods are paramount

-Who has it?

-What are their similarities?

-What are their differences





CHALLENGES & OPPORTUNITIES

- More data and technology at our hands then ever before – but maybe not for many people
- Previously expensive and exclusive tech has become widespread
- Accelerometery now wrist-worn by the majority
- Do we know why are we collecting it?
- Do we answer a specific and useful question with this?
- Do we store and retain aligned to GDPR and with appropriate consent? Do players' feel they have choice?

CHALLENGES & OPPORTUNITIES

Lack of cohesion

Data quality

Environment (funding; projects; short-term)

Politics

Innovation/hype

Significant technological advancements

Resources

Data literacy

Individual data

Open Science

'Life' prediction

Shared algorithms

ML & AI

Privacy – GDPR Compliance

CHALLENGES

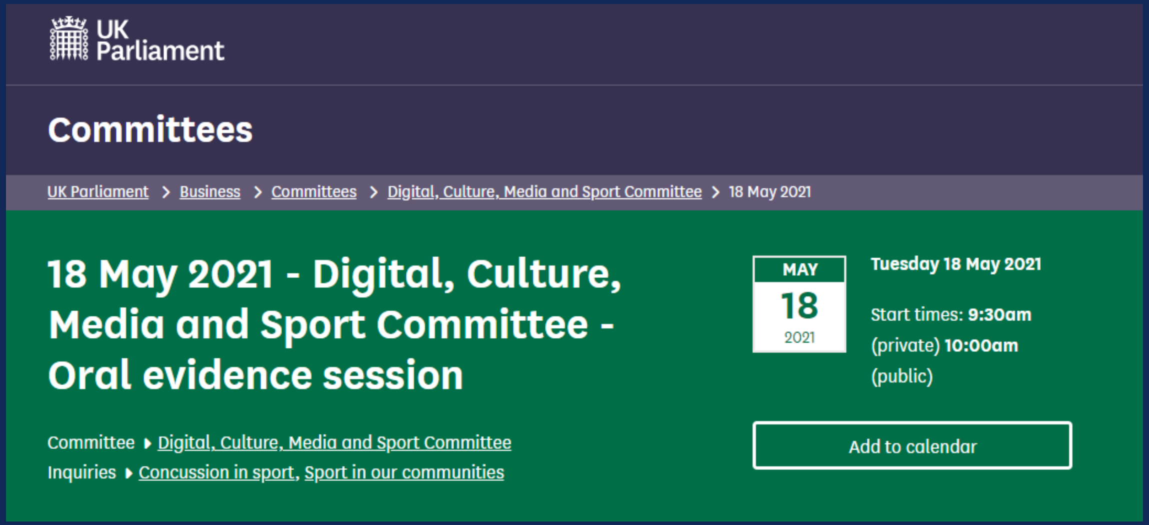
OPPORTUNITIES

WHAT'S NEXT

CHALLENGES & OPPORTUNITIES

FAST FORWARD… 166 YEARS

- There is drive for high-quality, robust data collection analysis, risk management and risk mitigation in medical environment
- This has vast ramifications, for patients and health services
- We are just hitting the tup of the iceberg



PART II

KEY CONCEPTS – AN EXAMPLE USING INJURY

- Self-report
- Medical attention
- Time loss (24h; 7-day; 28 day)
- Career ending/catastrophic

Severity

RTP

Exposure

Time at risk

- Recurrent?
- Subsequent?

KEY CONCEPT 1: DEFINE; CLASSIFY

- This can seem simplistic
- However, have you tried to run data collection for an injury project?
  - What did you include and exclude? Why? Did it work as you planned?
- If you have not; have you ever tried to compare injury types or incidence in a population? Did existing research provide you a clear message?
- Specific outcome, robust collection methodology

KEY CONCEPT 1: DEFINE; CLASSIFY

Theoretically

- An injury relates to physical damage to the body's tissues as a consequence of energy transfer and develops over a short period of time (Fuller, 2010)

- So how do we detect 'damage to the body's tissues' in a non-invasive, consistent and comparative way?

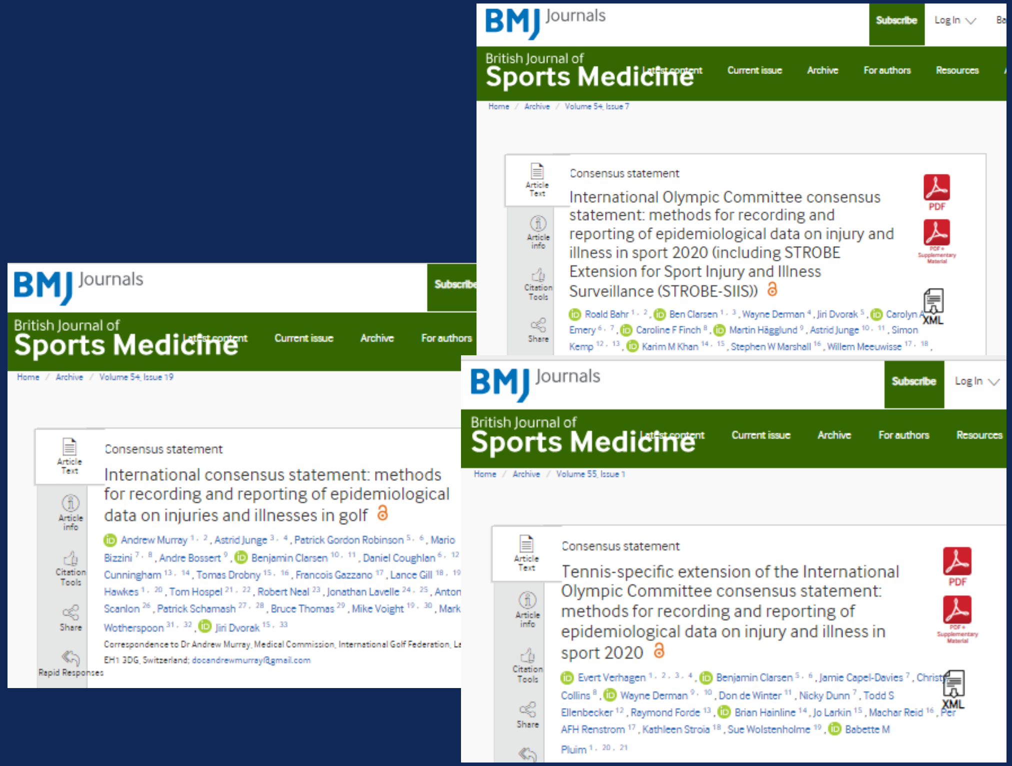
KEY CONCEPT 1: DEFINE; CLASSIFY

Operational definition

- Consensus achieved by academics and clinicians

- Sport-specific extensions *where needed*

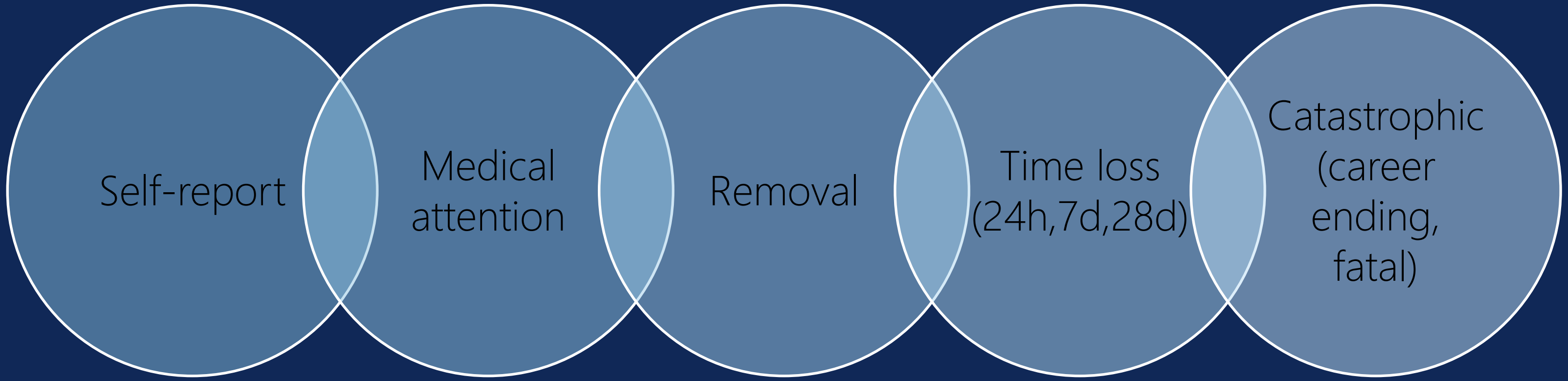
> Tissue damage or other derangement of normal physical function due to participation in sports, resulting from rapid or repetitive transfer of kinetic energy



KEY CONCEPT 1: DEFINE; CLASSIFY

NEXT

Given best practice (consensus), what is YOUR case definition, relevant to the population, setting, and research question



Most commonly linked to severity

KEY CONCEPT 1: DEFINE; CLASSIFY

Body region & structure affected; nature of injury

- International Classification of Diseases – World Health Organisation
- Occupational Injury & Illness Classification – US Dept of Labour
- → The Orchard Sports Injury Classification System (OSIICS)

Body Region

Main Group	Sub-group	OSICS (V8)
Head, neck	Head, face Neck, cervical spine	H N
Upper limb	Shoulder, clavicle Upper arm Elbow Forearm Wrist Hand, fingers	S U E R W P
Trunk	…	…
Lower limb	…	…

Structure Injured

Main Group	Sub-group	OSICS (V8)
Bone	Fracture Other bone injuries	F G, Q, S
Joint (non-bone), ligament	Dislocation, subluxation Sprain, ligament injury Lesion of meniscus, cartilage, disc	D, U J, L C
Muscle, tendon	…	…
Skin	…	…
Central & peripheral nervous systems	…	…
Other	Dental, Visceral, Other	G O

KEY CONCEPT 2: EXPOSURE

To derive a meaningful metric, the number of cases in isolation may not mean much

How many people were at risk? How long for?

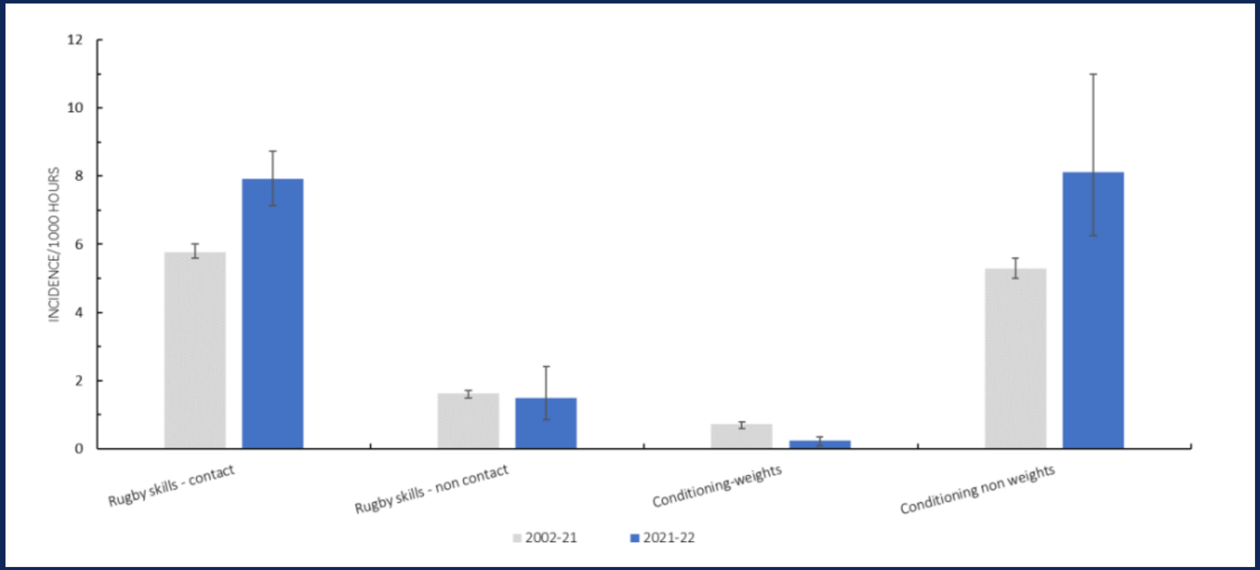
➔ If I run for 7hours but have no injury, and play ice hockey for 5 mins and sustain an injury, was my 'risk' in each activity equal?

KEY CONCEPT 2: EXPOSURE

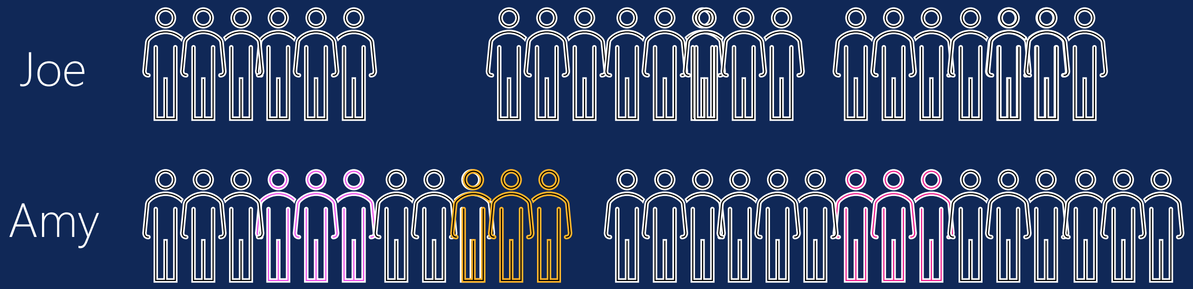
1) Is all exposure equivalent? Or do some periods need to be identified separately (match; group training and run-throughs; individual training/gym work)

   Are some environments more 'closed'?

2) Can I capture all exposure for my patient correctly? Do they play for other teams/clubs/representative sport?



Incidence of training injury types for 2021-22 vs 2002-21. Rugby Football Union's [PRISP report 2021-22]



KEY CONCEPT 3: STUDY DESIGN, RESEARCH QUESTION, SOURCES OF BIAS

- Set PICO: Population; Intervention; Comparison; Outcome;
- Choice of reference group is key
- Can you guarantee your 'non-exposed' group were actually not exposed
- If I compare a group with and without smoking in their likelihood of sustaining lung cancer, but half of my non-smoking group have smoking partners?
    - Inclusion and exclusion criteria
    - Selection bias
- Your non-exposed group should be alike your exposed group in every other facet – age, sex, comorbidities – balanced

KEY CONCEPT 4: CONFOUNDING

A → B

Time of day (AM/PM)

Deadlift (kg)

Nutrition

Age

Music

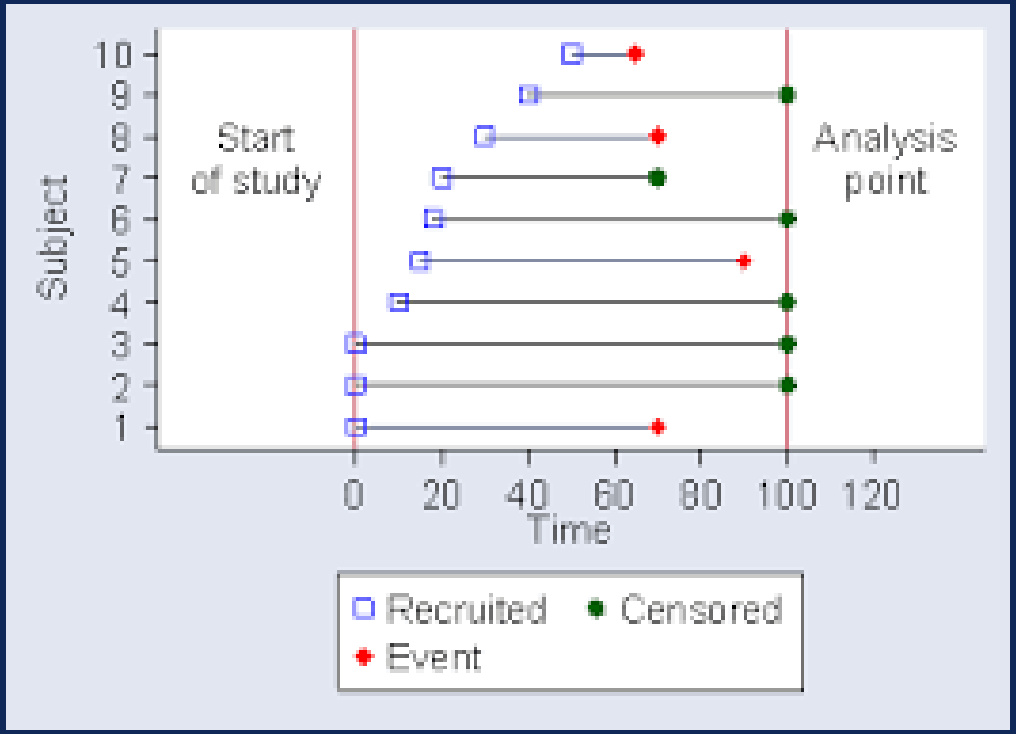
Training history

Sex

Your turn: Your team loses every away game…. The association between venue and match outcome?

METRICS

- Person Time
  - The number of exposures for all involved patients
  - Joe: 1 Day; Susie: 2 Days: Total Person Time = 3 Days
- Risk
  - Average probability of incurring the disease per patient
  - Risk of the disease is 20%.
- Censor
  - Patient occurrence of disease does not happen during observation
  - Right censor: Follow up ends without disease
  - Left Censor: Disease happens before observation begins



METRICS

- Incidence

  The number of cases of disease with their onset during a prescribed period of time

- Rate

  The number of cases of disease with their onset per unit of person time

  - 1 injury occurs in every 2.5 player-matches

- Prevalence

  The number of cases with a disease or attribute (i.e., injury) in a specific population

  - 30% of players are injured at any one time

Per 1000 AE? Per 1000 player-hours/match-hours???

Adapted from the Public Health Agency (2020). Incidence vs prevalence and the epidemiologist's bathtub | HSC Public Health Agency (hscni.net)

CAVEATS TO CONSIDER WHEN DESIGNING YOUR STUDY

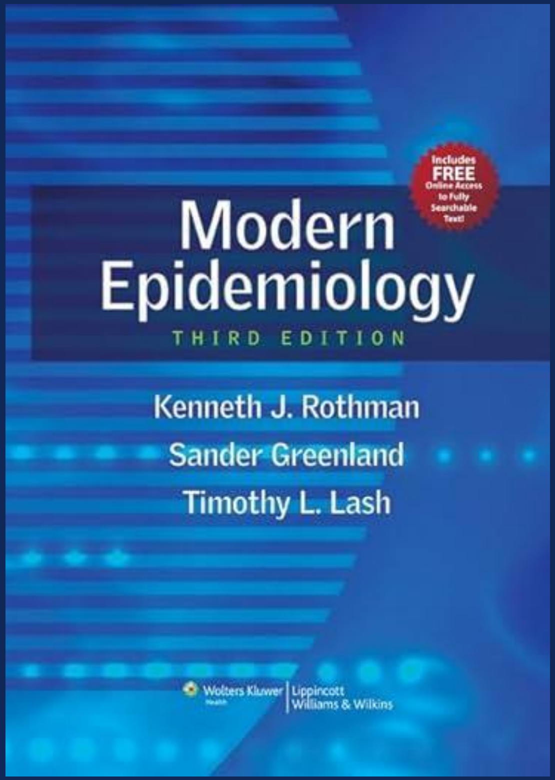
- Atypically healthy
- Low morbidity
- (Broadly) better health behaviours
- Poor recovery – stress
- Contractual ramifications and transparency
- Survival

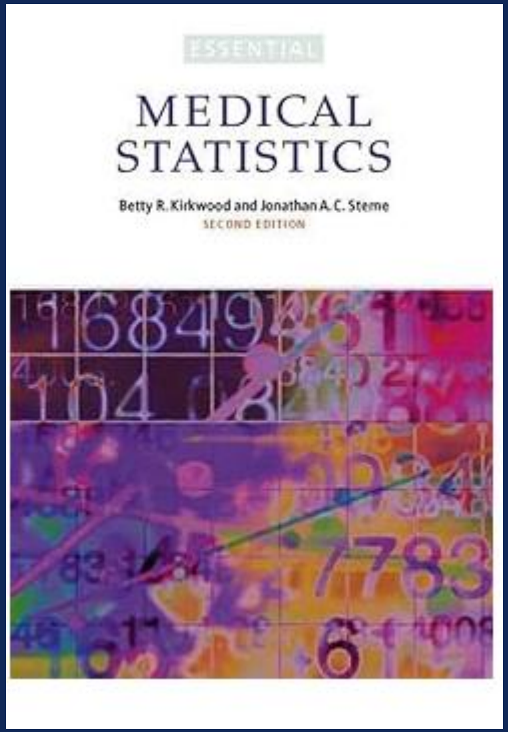
CAUTIONS

- Biological plausibility – is what we are finding actually likely? Or methodologically flawed
- Confounding, overfitting, evaluating
- Strong methodological underpinning and discussion, reflection should stay central in analysis
- Be your own peer-reviewer
- Stay current on GDPR, stay mindful of what you are storing and (appropriately) deleting – and leaving

RESOURCES

- IOC Injury and Illness Consensus – Currently International Olympic Committee consensus statement: methods for recording and reporting of epidemiological data on injury and illness in sport 2020 (including STROBE Extension for Sport Injury and Illness Surveillance (STROBE-SIIS)) | British Journal of Sports Medicine (bmj.com)





IN CLASS ACTIVITY

- The Dean of the School of Medicine has come to you as the biomedical data scientist with a problem to solve!
- Hospital readmissions following knee arthroplasty seem to be increasing over the last quarter, with no explanation why.
- How would you design a prospective study to evaluate this?
- Remember PICO here.

FINAL THOUGHTS

- Think critically of study design
  - Exact population
  - Precise outcome
  - What's the comparison
  - Confounding and common causes
- What epidemiological calculations (prevalence, incidence, rate, etc.) inform your problem